

Electrostatic Potential & capacitance

Electric potential Energy difference

→ Work done is not defined at a point it must be b/w two point.



$$-W_{c.f} = \Delta U = U_B - U_A$$

$$\Delta U = -W_{c.f} \rightarrow \text{Always valid}$$

$$-\frac{W_{c.f}}{q} = \frac{\Delta U}{q} = \Delta V \rightarrow \Delta V = \frac{\Delta U}{q}$$

$$\Delta U = q \Delta V$$

$$-W_{c.f} = \Delta U = q \Delta V$$

Electric potential

→ Work done in bringing unit 've' charge from infinity to the point without change in K.E is called Electric potential.

→ -ve work done in bringing unit 've' charge from '∞' to the point.

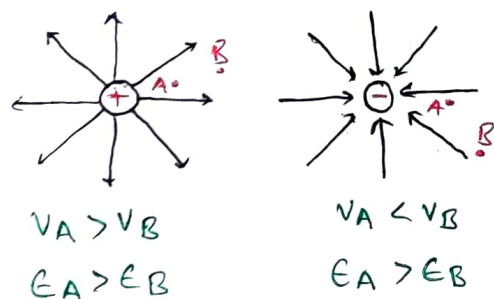
$$\Delta V = - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{s} = - \int_{r_A}^{r_B} \frac{F_{c.f}}{q} \cdot dr = - \frac{W_{c.f}}{q}$$

$$\# E = - \frac{dV}{dr} = - [\text{slope of } V-r \text{ graph}]$$

$$E = - \left[\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right]$$

⇒ Super Note:-

- At a point, Electric potential does not have unique value.
 - Electric potential depends on reference point, but ΔV does not.
 - Scalar physical quantity
 - Unit → volt, J/C, Weber/sec, N·m/C.
- Potential decreases in the dirⁿ of E.f.

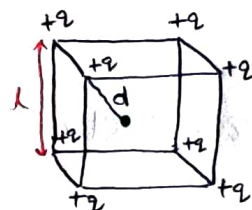


Q → Assertion-Reason type Questions

- ① When work done by C.f is -ve, then potential/P.E will increase → **True**.
- ② When work done by C.f is +ve, the potential must decrease → **True**.
- ③ Can +ve charge, create -ve potential? → **No**.
- ④ If potential is constant then electric field must be zero → **True**.
- ⑤ Electric field is zero, then potential may be non-zero → **True**.

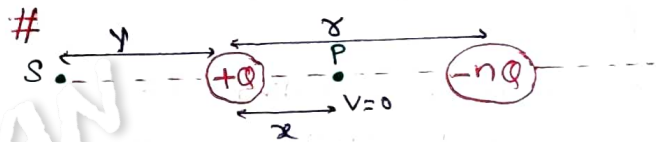
Electric potential due to combination of charge

Q → find Electric potential at centre of cube



$$V = \frac{8Kq}{\frac{\sqrt{3}l}{2}}$$

$$V = \frac{16Kq}{\sqrt{3}l}$$



$$y = (n-1)x$$

$$y = nx + x$$

$$y = \left(\frac{x}{n-1} \right)$$

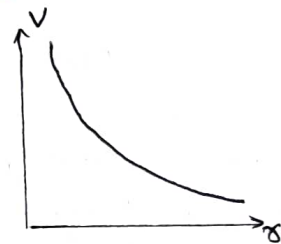
$$x = \frac{y}{(n+1)}$$

Potential due to:-

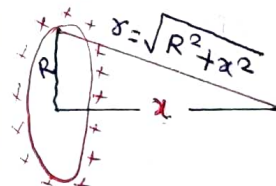
① Point charge:-

$$\oplus V_p = \frac{KQ}{r}$$

$$\ominus V_p = -\frac{KQ}{r}$$



② Ring:-



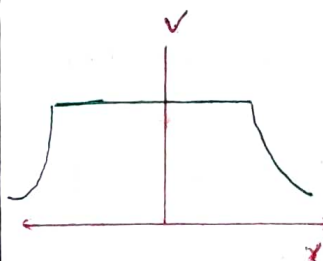
$$V = \frac{KQ}{\sqrt{R^2 + x^2}}$$

At centre (x=0)

$$V_c = \frac{KQ}{R}$$

If $x \gg R$

$$V = \frac{KQ}{x}$$



③ Halb-Ring:-

$$V_0 = \frac{KQ}{R}$$



④ Circular arc:-

$$V_0 = \frac{KQ}{R}$$

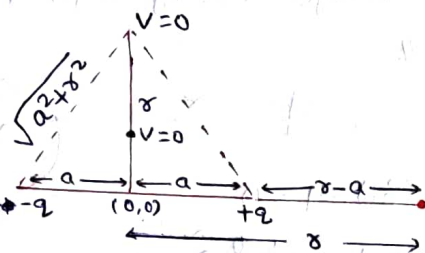


⑤ Dipole:-

$$\vec{P} = (q2a)\hat{i}$$

$$V = \frac{2Kqa}{r^2 - a^2} = \frac{KP}{r^2 - a^2}$$

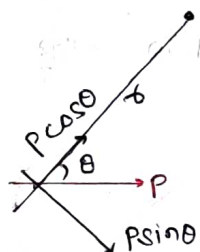
Small dipole ($r \gg a$)



$$V = \frac{KP}{r^2}$$

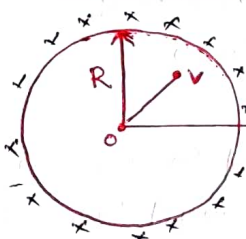
$$V_{\text{equatorial}} = 0$$

⑥ Dipole at angle 'θ'



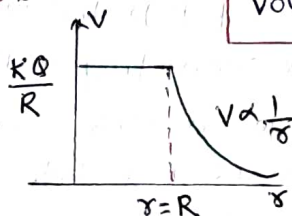
$$V = \frac{K(P \cos \theta)}{r^2}$$

⑦ Hollow (conducting / non-conducting) or solid conducting sphere:-

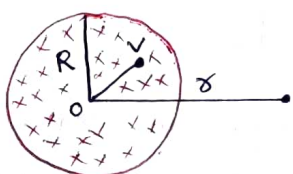


$$V_{\text{cent}} = V_{\text{inside}} = V_{\text{surf}} = \frac{KQ}{R}$$

$$V_{\text{out}} = \frac{KQ}{r}$$



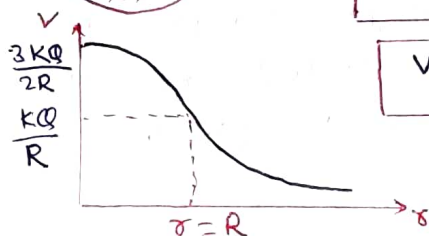
⑧ Solid non-conducting sphere



$$V_{\text{inside}} = \frac{KQ(3R^2 - r^2)}{2R^3}$$

$$V_{\text{surf}} = \frac{KQ}{R} \quad V_{\text{cent}} = \frac{3KQ}{2R}$$

$$V_{\text{out}} = \frac{KQ}{r}$$



⑨ Due to infinite thin charge sheet



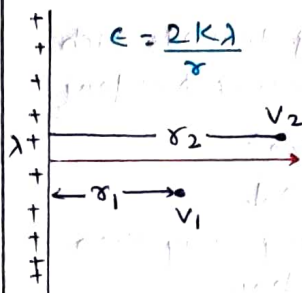
$$\epsilon = \frac{\sigma}{2\epsilon_0}$$

$$V_1 - V_2 = \epsilon \cdot r$$

$$V_1 - V_2 = \frac{\sigma r}{2\epsilon_0}$$

$$V_1 - V_2 = \frac{\sigma(r_2 - r_1)}{2\epsilon_0}$$

⑩ Due to infinite line charge density λ



$$\epsilon = \frac{2K\lambda}{r}$$

$$\Delta V = -\epsilon dr$$

$$V_2 - V_1 = -2K\lambda \log\left(\frac{r_2}{r_1}\right)$$

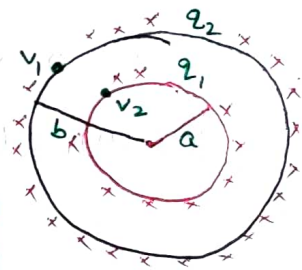
$$V_1 - V_2 = 2K\lambda \log\left(\frac{r_2}{r_1}\right)$$

Electric potential due to group of concentric shells

→ potential diff b/w surface of two spheres

$$V_1 = \frac{Kq_2}{b} + \frac{Kq_1}{b}$$

$$V_2 = \frac{Kq_1}{a} + \frac{Kq_2}{b}$$



$$\Delta V = V_1 - V_2 = Kq_1\left(\frac{1}{b} - \frac{1}{a}\right)$$

↓ doesn't depend on charge of outer sphere.

Equipotential surface

→ An equipotential surface is a surface with a constant potential at all points of surface.

- Electric field is perpendicular to its surface.
- Along equipotential surface, electric field must be zero.

$$\rightarrow \Delta V = 0, \quad w = 0$$

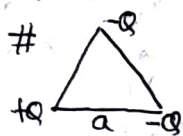
- for point charge → spherical
- for line charge → cylindrical
- for charge sheet → sheet
- At large distance from ring or any charge distribution → spherical
- Infinite charge plate → plate

Potential Energy of a system of charges

→ Potential energy of two point charge



$$U = \frac{kq_1q_2}{r}$$



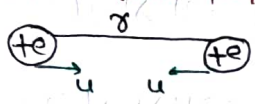
$$U = -\frac{kq^2}{a} + \frac{kq^2}{a} - \frac{kq^2}{a}$$

$$U = -\frac{kq^2}{a} \text{ Ans}$$

→ Total no. of terms of potential energy

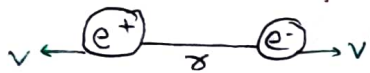
$$\text{for } n\text{-charge} = \frac{N(N-1)}{2}$$

Two proton is moving towards each other then potential energy of system



$$\uparrow U = \frac{ke^2}{r} \downarrow$$

One proton & e⁻ moving away then potential energy of system.



$$\uparrow U = -\frac{ke^2}{r} \uparrow$$

→ Conservation of Mechanical Energy

If $W_{ext} = 0$, Apply C.O.M.E

$$(K.E + U)_i = (K.E + U)_f$$

Self Energy

① Hollow conducting / non-conducting or solid conducting sphere:-

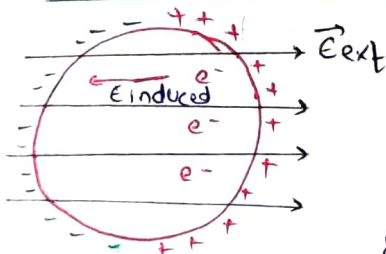
$$U = \frac{kq^2}{2R}$$

② Solid non-conducting sphere:-

$$U = \underbrace{0.5 \frac{kq^2}{R}}_{\text{outside}} + \underbrace{0.1 \frac{kq^2}{R}}_{\text{Inside}}$$

$$U = \frac{0.6kq^2}{R} = \frac{3kq^2}{5R}$$

Electrostatics of Conductor



$$E_{net} = E_{ext} + E_{ind}$$

$$E_{net} = 0$$

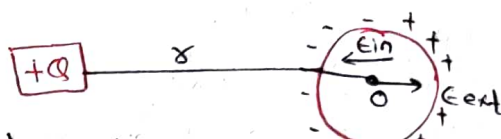
$$E_{ext} = E_0$$

$$E_{induced} = -E_0$$

NOTE:-

- Electric field inside conductor is zero, where conducting matter is present.
- Electric field always $\perp$ to surface of conductor.
- $E_{(surface)} = \frac{\sigma}{\epsilon_0}$.
- When two conductor is connected, then their potential becomes same.
- Conductor creates equipotential region.
- charge density high at sharp point.

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$$(E_0)_{net} = 0$$

$$(E_0)_{+Q} = \frac{kQ}{r^2} \hat{r}$$

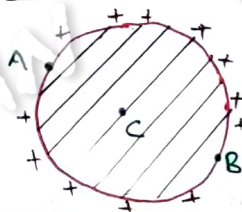
$$(E_0)_{\text{due induced}} = -\frac{kQ}{r^2} \hat{r}$$

neutral solid Conductor

→ Excess charge given to conductor

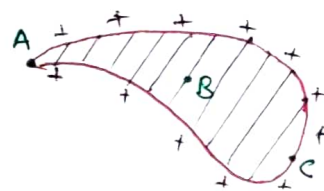
(i) Uniform shape

(ii) Random shape



$$V_A = V_B = V_C$$

$$E_A = E_B, E_C = 0$$



$$V_A = V_B = V_C$$

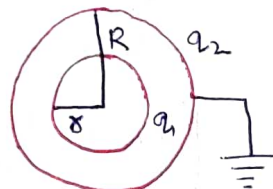
$$E_A > E_C > E_B$$

$$E_B = 0$$

Earthing of Conductor

→ charge may or may not be zero on conductor but potential must become zero after connecting with earth.

→ Final charge on each sphere when outer sphere is grounded.



→ Jisko ground nahi kiya hai uska charge same rahega, jisko ground kiya hai, uska potential zero hoga & final charge on outer sphere.

CAPACITOR

- It is an electrical device which store electrostatic energy by storing charge.
- Combination of two conductors, having equal & opposite charge. ($Q = CV$)

$$C = \frac{Q}{\Delta V} = \frac{Q}{V_1 - V_2}$$



↳ Scales

↳ Unit (C/V) = farad

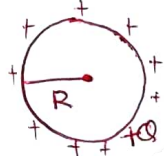
↳ Dimension: $C = \frac{Q}{V} = \frac{C}{J/C} = \frac{C^2}{J} = \frac{I^2 T^2}{M L^2 T^{-2}} = \frac{I^2 T^2}{M L^2}$

↳ Air is insulator upto certain field, after that it behave as conductor, that limit is called dielectric strength of air ($3 \times 10^6 \text{ N/C}$).

Capacitance of an isolated spherical conductor.

$$C = \frac{Q}{V} = \frac{Q}{\frac{kQ}{R}} = \frac{R}{k}$$

$$C = 4\pi\epsilon_0 R$$



⇒ 'n' small charged sphere combine to form a bigger sphere.



$$C = 4\pi\epsilon_0 R$$

$$C \propto R \Rightarrow C = C_0 n^{1/3}$$

$$R = n^{1/3} r$$

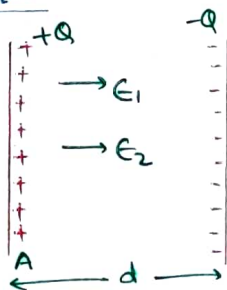
Parallel plate capacitor

$$C_{\text{net}} = C_1 + C_2 = \frac{Q}{2A\epsilon_0} + \frac{Q}{2A\epsilon_0}$$

$$C_{\text{net}} = \frac{Q}{A\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

$$\Delta V = E \cdot d$$

$$\Delta V = \frac{Qd}{A\epsilon_0}$$



$$C_0 = \frac{Q}{\Delta V} = \frac{A\epsilon_0}{d}$$

$$C_{\text{med}} = \frac{E \cdot A}{d} = K C_0$$

Force between the plates

$$F = QE$$

$$F = \frac{Q^2}{2A\epsilon_0} = \frac{\sigma^2 A}{2\epsilon_0}$$

pressure on the plate

$$P = \frac{F}{A} = \frac{Q^2}{2A^2\epsilon_0} = \frac{\sigma^2}{2\epsilon_0}$$

Energy density (U)

$$U = \frac{E}{V}$$

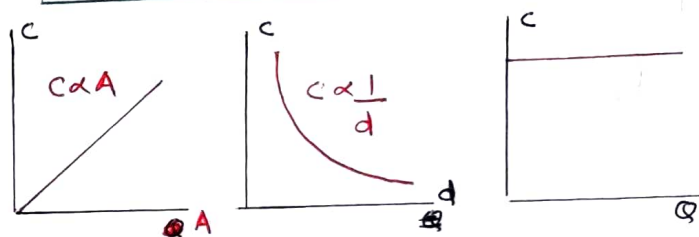
$$U = \frac{\sigma^2}{2\epsilon_0} = \frac{1}{2} \epsilon_0 E^2 = \frac{Q^2}{2A^2\epsilon_0}$$

$$\frac{E}{Ad} = \frac{Q^2}{A^2 2\epsilon_0}$$

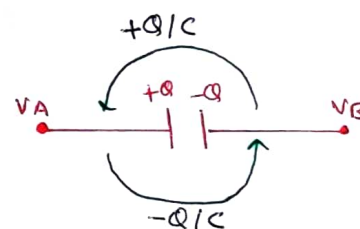
$$Q = CV$$

$$E = \frac{Q^2 d}{2\epsilon_0 A} = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{QV}{2}$$

Graph for parallel plate capacitor



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A to B

$$V_A - Q/C = V_B$$

$$V_A - V_B = Q/C$$

B to A

$$V_B + Q/C = V_A$$

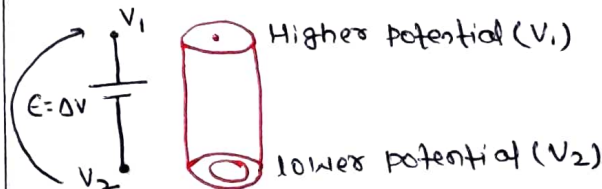
$$V_A - V_B = Q/C$$

BATTERY

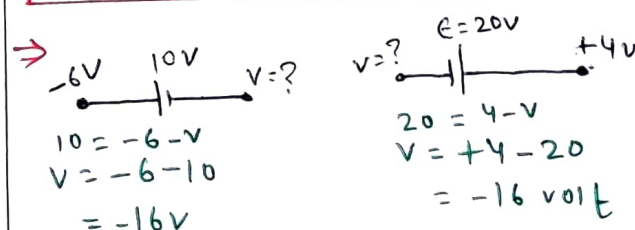
↳ Source of Energy not a source of charge.

↳ It maintain potential difference b/w the points where it is connected.

↳ Work done by battery = $Q\Delta V$



$$E = \Delta V = (V_1 - V_2) = e \cdot m \cdot f$$



$$10 = -6 - V$$

$$V = -6 - 10$$

$$= -16V$$

$$20 = 4 - V$$

$$V = +4 - 20$$

$$= -16 \text{ volt}$$

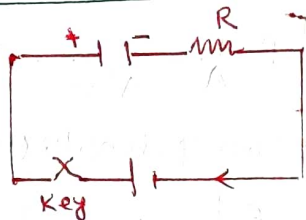
charging of capacitor

$$W_{\text{Battery}} = qV = cv(v) = cv^2$$

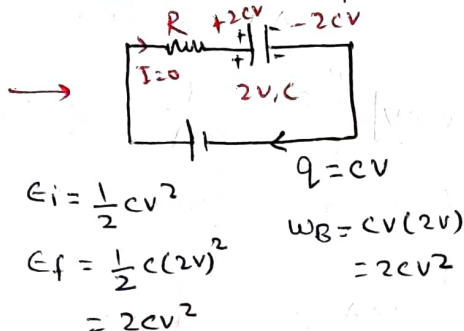
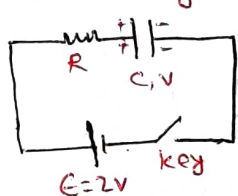
$$E_{\text{store}} = \frac{1}{2} cv^2$$

$$E_{\text{loss}} = \frac{1}{2} cv^2$$

$$E_i + W_B = E_f + E_{\text{loss}}$$



When key is closed find Energy loss.



$$\Rightarrow \frac{1}{2} cv^2 + 2cv^2 = 2cv^2 + E_{\text{loss}}$$

$$E_{\text{loss}} = \frac{1}{2} cv^2$$

Combination of capacitors

(A) Series combination of capacitors

Q → Same on each capacitor

$$V = V_1 + V_2 + V_3 + \dots$$

$$\frac{1}{C_q} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

⇒ Law of potential drop:-

$$Q = CV = \text{same} \quad C \propto \frac{1}{V} \quad \frac{V_1}{V_2} = \frac{C_2}{C_1}$$

(B) Parallel combination of capacitors

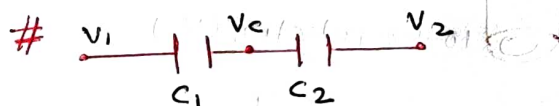
V → same

$$Q = Q_1 + Q_2 + Q_3 + \dots$$

$$C_{eq} = C_1 + C_2 + C_3 + \dots$$

Law of charge distribution

$$Q = CV \quad Q \propto C$$

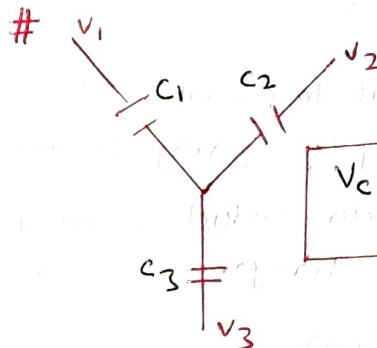


$$V_c = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

If $V_1 > V_2$

$$Q_1 = C_1 (V_1 - V_c)$$

$$Q_2 = C_2 (V_c - V_2)$$



$$V_c = \frac{C_1 V_1 + C_2 V_2 + C_3 V_3}{C_1 + C_2 + C_3}$$

⇒ If n-identical connected in series & Parallel:-

Series

$$C_{eq} = \frac{C}{n}$$

Parallel

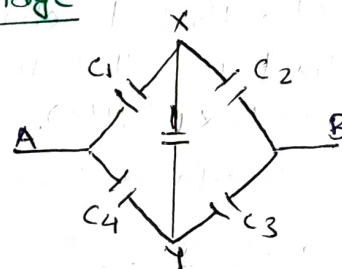
$$C_{eq} = nC$$

⇒ Wheat stone bridge

$$\# V_x = V_y$$

charge on C = 0

$$\# \frac{C_1}{C_4} = \frac{C_2}{C_3}$$



Series combination in Dielectrics

$$C = \frac{A\epsilon_0}{\left[\frac{d_1}{K_1} + \frac{d_2}{K_2} \right]}$$

$$K_{eq} = \frac{d}{\frac{d_1}{K_1} + \frac{d_2}{K_2}}$$

$$\text{If } d_1 = d_2 = \frac{d}{2}, K_{eq} = \frac{2K_1 K_2}{K_1 + K_2}$$

Parallel combination in dielectrics

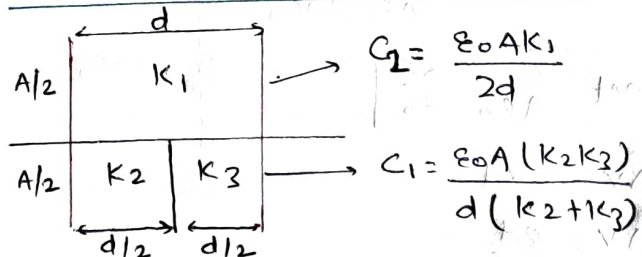
$$C_{eq} = \frac{\epsilon_0}{d} [K_1 A_1 + K_2 A_2 + \dots]$$

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$$

If $A_1 = A_2 = \frac{A}{2}$

$$K_{eq} = \frac{K_1 + K_2}{2}$$

⇒ Calculate capacitance b/w X & Y



$$C_2 = \frac{\epsilon_0 A K_1}{2d}$$

$$C_1 = \frac{\epsilon_0 A (K_2 K_3)}{d(K_2 + K_3)}$$

$$C_{eq} = C_1 + C_2$$

$$C_{eq} = \frac{\epsilon_0 A}{d} \left[\frac{K_2 K_3}{K_2 + K_3} + \frac{K_1}{2} \right]$$

If $K_1 = K_2 = K_3 = 1$

$$C = \frac{\epsilon_0 A}{d}$$

Capacitor of capacitance C_0 is charged by a battery of e.m.f V_0 , Now battery remains connected & dielectric plate placed inside it.

	Before	After
Capacitance	C_0	$K C_0$
Potential	V_0	V_0 (same)
charge ($Q = CV$)	Q_0	$K Q_0$
Electric field	E_0	E_0 (same)
Energy	U_0	$K U_0$
force on plate	$f_0 = QE$	$K f_0$

Battery removed then dielectric is placed :-

	Before	After
Capacitance	C_0	$K C_0$
Potential	V_0	V_0 / K
Charge	Q_0	Q_0 (same)
Electric field	E_0	E_0 / K
Energy	U_0	U_0 / K
force on plate	f_0	$\frac{f_0}{K}$